5.5 find $var(b_2|X)$ and examine $\lim\limits_{T\to\infty} var(b_2|X) = 0$

(a) $$Var(b_2|X) = \frac{\sigma^2}{\sum\limits_{t=1}^{T}(x_t-\bar{x})^2} = \frac{\sigma^2}{\sum\limits_{t=1}^{T} x_i^2 - \frac{(\sum x_i)^2}{T}}$$

let $x_t = t$, where $t = 1, 2, 3, \dots, T$

$$\therefore \sum_{t=1}^{T} x_i^2 - \frac{(\sum(x_i))^2}{T} = \sum_{t=1}^{T} t^2 - \frac{(\sum_{t=1}^{T} t)^2}{T}$$

$$= \frac{T(T+1)(2T+1)}{6} - \frac{1}{T}\times\left(\frac{T(T+1)}{2}\right)^2 = \frac{(T^2+T)(2T+1)}{6} - \frac{T(T+1)^2}{4}$$

$$= \frac{(2T^2+2T)(2T+1)-3T(T+1)^2}{12} = \frac{4T^3+\cancel{6T^2}+2T-3T^3-\cancel{6T^2}-3T}{12} = \frac{T^3-T}{12}$$

$$\Rightarrow Var(b_2|X) = \frac{12\sigma^2}{T(T+1)(T-1)} \text{ and } \lim_{T\to\infty} Var(b_2|x) = 0$$

故 the LSE estimator b_2 is consistent #

(b) let $x_t = 0.5^t$ $\therefore$ $$Var(b_2|X) = \frac{\sigma^2}{\sum\limits_{t=1}^{T}(0.5^t-\bar{x})^2}$$

$$\sum_{t=1}^{T}(0.5^t-\bar{x})^2 = \sum_{t=1}^{T} 0.5^{2t} - \underbrace{2\times\bar{x}\times T\times\bar{x}}_{=-2\bar{x}^2T} + T\bar{x}^2 = \sum_{t=1}^{T} 0.5^{2t} - \bar{x}^2T$$

$$= \sum_{t=1}^{T} 0.25^t - \frac{1}{T}\left(\sum_{t=1}^{T} 0.5^t\right)^2 = \frac{0.25(1-0.25^T)}{1-0.25} - \frac{1}{T}\times\left(\frac{0.5(1-0.5^T)}{1-0.5}\right)^2$$

$$= \frac{1}{3}(1-0.25^T) - \frac{1}{T}(1-0.5^T)^2$$

故 $$Var(b_2|X) = \frac{\sigma^2}{\frac{1}{3}(1-0.25^T) - \frac{1}{T}(1-0.5^T)^2}$$

and $$\lim_{T\to\infty} Var(b_2|X) = \frac{\sigma^2}{\frac{1}{3}\times 1 - 0} = \frac{\sigma^2}{3} \neq 0$$

$\therefore$ the LSE estimator b_2 is not consistent. #

(c) in (a). when T increases, it will provide more information about β_2 and the relationship between x and y. Nevertheless, in (b), $x = 0.5^t$ becomes zero for large enough sized T, and increasing T further does not provide any more infomation about β_2 #

5.6 $N=63$, $df=N-K=60$

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix}, \quad \widehat{cov}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}, \quad \alpha = 0.05$$

a. $H_0: \beta_2 = 0$, $H_a: \beta_2 \neq 0$

$$t_{STAT} = \frac{3-0}{\sqrt{4}} = 1.5$$

又 $t_c = t_{df=60,\, 0.975} = 2$

$\because t_{STAT} = 1.5 < t_c = 2$ ($p\text{-value} > \alpha = 0.05$)

$\therefore$ 不拒絕 $H_0: \beta_2 = 0$ #

b. $H_0: \beta_1 + 2\beta_2 = 5$, $H_a: \beta_2 + 2\beta_2 \neq 5$

$$t_{STAT} = \frac{b_1 + 2b_2 - 5}{se(b_1+2b_2)} = \frac{3}{\sqrt{11}} \fallingdotseq 0.9045$$

代入

$var(b_1 + 2b_2) = var(b_1) + 4var(b_2) + 4cov(b_1, b_2) = 3 + 16 - 8 = 11$

$\therefore se(b_1 + 2b_2) = \sqrt{11}$

$\because t_{STAT} = 0.9045 < t_c = 2$ ($p\text{-value} > \alpha = 0.05$)

$\therefore$ 不拒絕 H_0 #

c、 $H_0: \beta_1 - \beta_2 + \beta_3 = 4$, $H_a: \beta_1 - \beta_2 + \beta_3 \neq 4$

$$t_{STAT} = \frac{b_1 - b_2 + b_3 - 4}{se(b_1 - b_2 + b_3)} = \frac{-6}{4} = -1.5$$

$$var(b_1 - b_2 + b_3) = var(b_1) + var(b_2) + var(b_3) - 2Cov(b_1, b_2) - 2Cov(b_2, b_3) + 2Cov(b_1, b_3)$$

$$= 3 + 4 + 3 + 4 + 0 + 2 = 16$$

代入

$$\Rightarrow se(b_1 - b_2 + b_3) = 4$$

$\because t_{STAT} = -1.5 > -t_c = -2$ (p-value $> \alpha = 0.05$)

$\therefore$ 拒絕 $H_0: \beta_1 - \beta_2 + \beta_3 = 4$ #

5. 20

(a) 將 $(\bar{x}_1, \bar{y}_1)$, $(\bar{x}_2, \bar{y}_2)$ 代入 regression model

$\bar{y}_1 = \beta_1 + \beta_2 \bar{x}_1 + \frac{1}{3}\sum_{i=1}^{3} e_i$, $\bar{y}_2 = \beta_1 + \beta_2 \bar{x}_2 + \frac{1}{3}\sum_{i=4}^{6} e_i$, and let

$\frac{1}{3}\sum_{i=1}^{3} e_i = \bar{e}_1$, $\frac{1}{3}\sum_{i=4}^{6} e_i = \bar{e}_2$

$$\Rightarrow \hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1} = \frac{(\cancel{\beta_1} + \beta_2\bar{x}_2 + \bar{e}_2) - (\cancel{\beta_1} + \beta_2\bar{x}_1 + \bar{e}_1)}{\bar{x}_2 - \bar{x}_1}$$

$$= \frac{\beta_2(\bar{x}_2 - \bar{x}_1) + (\bar{e}_2 - \bar{e}_1)}{\bar{x}_2 - \bar{x}_1} = \beta_2 + \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1}$$

$$\because E(\hat{\beta}_{2,mean} \mid X) = \beta_2 + \frac{1}{\bar{x}_2 - \bar{x}_1} E(\underbrace{\bar{e}_2 - \bar{e}_1}_{} \mid X) = 0 - 0$$

$$= E(\bar{e}_2 \mid X) - E(\bar{e}_1 \mid X) = 0 - 0 = 0$$

$$\Rightarrow E(\hat{\beta}_{2,mean} \mid X) = \beta_2$$

$$\therefore E(\hat{\beta}_{2,mean}) = E[E(\hat{\beta}_{2,mean} \mid X)] = E(\beta_2) = \beta_2$$

故 $\hat{\beta}_{2,mean}$ is an unbiased estimator of β_2 #

(b)

$$\because\ \hat{\beta}_{2,\text{mean}} - \beta_2 = \frac{\bar{e}_2 - \bar{e}_1}{\bar{x}_2 - \bar{x}_1} \Rightarrow Var(\hat{\beta}_{2,\text{mean}} | X) = \frac{Var(\bar{e}_2 - \bar{e}_1 | X)}{(\bar{x}_2 - \bar{x}_1)^2}$$

$$又\ Var(\bar{e}_2 - \bar{e}_1 | X) = Var(\bar{e}_2 | X) + Var(\bar{e}_1 | X)$$

$$= Var\left(\frac{1}{3}\sum_{i=4}^{6} e_i \,\middle|\, X\right) + Var\left(\frac{1}{3}\sum_{i=1}^{3} e_i \,\middle|\, X\right)$$

$$= \frac{1}{9} \times 3\sigma^2 + \frac{1}{9} \times 3\sigma^2 = \frac{2}{3}\sigma^2$$

$$\therefore\ Var(\hat{\beta}_{2,\text{mean}} | X) = \frac{2\sigma^2}{3(\bar{x}_2 - \bar{x}_1)^2}$$ #

(c) Note that: $Var(\hat{\beta}_{2,\text{mean}}) = E[Var(\hat{\beta}_{2,\text{mean}} | X)] + Var(E(\hat{\beta}_{2,\text{mean}} | X))$

$$\begin{cases} Var(E(\hat{\beta}_{2,\text{mean}} | X)) = Var(\beta_2) = 0 \\ E(Var(\hat{\beta}_{2,\text{mean}} | X)) = E\left[\dfrac{2\sigma^2}{3(\bar{x}_2 - \bar{x}_1)^2}\right] = \dfrac{2}{3}\sigma^2 E\left[\dfrac{1}{3(\bar{x}_2 - \bar{x}_1)^2}\right] \end{cases}$$

$$\therefore\ Var(\hat{\beta}_{2,\text{mean}}) = \frac{2}{3}\sigma^2 E\left[\frac{1}{\bar{x}_2 - \bar{x}_1}\right]$$ #

(d)

$$Var(\hat{\beta}_{2,mean}|X) = \frac{4\sigma^2}{N(\bar{X}_2-\bar{X}_1)^2} = \frac{4\sigma^2}{9N}$$

$\Rightarrow Var(\hat{\beta}_{2,mean}|X) \to 0$ as $N(\bar{X}_2-\bar{X}_1)^2 \to \infty$

As a result, $Var(\hat{\beta}_{2,mean}|X)$ is consistent if $\lim_{N\to\infty}(\bar{X}_2-\bar{X}_1)^2 N \to \infty$ ✗

(e) (f) 放最後面

(g)

① $E[(S_x^2)^{-1}] \xrightarrow{P} 0.12$

$\because X_i \sim U(0,10)$, iid and $Var(X) = \frac{(10-0)^2}{12} = \frac{100}{12}$

by 大數法則: $S_x^2 = \frac{1}{N-1}\left(\sum_{i=1}^{N}(X_i-\bar{X})^2\right) \xrightarrow{P} Var(X) = \frac{100}{12}$

$\because S_x^2 \xrightarrow{P} \frac{100}{12} \Rightarrow S_x^{-2} \xrightarrow{P} \left(\frac{100}{12}\right)^{-1} = 0.12$

and hence $E(S_x^{-2}) \xrightarrow{P} 0.12$

② $\frac{4}{(\bar{X}_2-\bar{X}_1)^2} \xrightarrow{p} 0.16$

note that 前半樣本集中於 $(0,5)$，而後半樣本集中於 $(5,10)$

and $E(X \mid X \in (0,5)) = \frac{5+0}{2} = 2.5$, $E(X \mid X \in (5,10)) = \frac{5+10}{2} = 7.5$

Hence when $N \to \infty \Rightarrow \bar{X}_1 \xrightarrow{p} 2.5$, $\bar{X}_1 \xrightarrow{p} 7.5 \Rightarrow \bar{X}_2 - \bar{X}_1 \xrightarrow{p} 5$

and $\frac{4}{(\bar{X}_2-\bar{X}_1)^2} \xrightarrow{p} \frac{4}{\sigma^2} = 0.16$, hence $E\left(\frac{4}{(\bar{X}_2-\bar{X}_1)^2}\right) = 0.16$ *

(h)

$\hat{\beta}_{2,mean}$ will not be consistent.

∵ for $\bar{X}_1, \bar{X}_2$ ，都會 coverage in probability to 5 (皆為 random sample) $\Rightarrow \bar{X}_2 - \bar{X}_1 \xrightarrow{p} 0$

∴ $Var(\hat{\beta}_{2,mean} \mid x) = \frac{4\sigma^2}{N(\bar{X}_2-\bar{X}_1)^2}$ 無法確認是否收斂到0 *

(e) i. ii.

1	100	1.24362517	1.6631963	0.1243625	0.1663196
2	500	0.24402399	0.3260404	0.1220120	0.1630202
3	1000	0.12290628	0.1656282	0.1229063	0.1656282
4	5000	0.02413174	0.0323161	0.1206587	0.1615805

$\downarrow$ $Var(b_2|X)$ $\quad\downarrow$ $Var(\hat{\beta}_{2,mean}|X)$ $\quad\downarrow$ $E[(S_x^2)^{-1}]$ $\quad\downarrow$ $E\left[\frac{4}{(\bar{x}_2-\bar{x}_1)^2}\right]$

(f)

① by the result of (e),

$$\begin{cases} Var(b_2|X) = \dfrac{\sigma^2}{N S_x^2}, \quad Var(\hat{\beta}_{2,mean}|X) = \dfrac{4\sigma^2}{N(\bar{x}_2-\bar{x}_1)^2} \\ E(S_x^{-2}) \doteq 0.12, \quad E\left[\dfrac{4}{(\bar{x}_2-\bar{x}_1)^2}\right] \doteq 0.16 \end{cases}$$

Hence $E[Var(b_2|X)] = \frac{\sigma^2}{N} E\left[\frac{4}{(\bar{x}_2-\bar{x}_1)^2}\right] \approx \frac{0.16}{N}\sigma^2$

$\therefore Var(\hat{\beta}_{2,mean}) > Var(b_2)$

② 當N越大，二者分別下降且趨近於 0.12 & 0.16

③ $\because X_i \sim U(0,10) \Rightarrow$ when $N \to \infty$, $\bar{x}_1 \to 2.5$, $\bar{x}_2 \to 7.5$

$\therefore Var(\hat{\beta}_{2,mean}|X) \to \dfrac{4\sigma^2}{(7.5-2.5)N} = \dfrac{0.16}{N}\sigma^2 \to 0$ as $N \to \infty$ #